

Anti-biofouling potential of extremely water-repellent carbon soot coatings immersed in a highly-contaminated seawater swamp

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ABSTRACT

The undesired translocation of marine microorganisms during sailing and the excessive expenditures for ship hull repairment due to biofouling could be circumvented using superhydrophobic coatings. Unfortunately, the combined effect of underwater currents and colonizing activity of multitudinous foulants on the anti-microbial performance of existing non-wettable materials is still at infant stage. The paramount objective here is to evaluate for the first time the potential of two mechanically durable water-repellent soot coatings towards the inhibition of bacterial proliferation in "real-life" outdoor conditions. We demonstrate that upon one-week immersion in a seawater swamp, both specimens lose non-wettability due to dissolution of the trapped gas layer into the water basin, but yet manage to retard the biofilm development, compared to uncoated iron surfaces, and halt the settlement of some Gram-negative and Gram-positive bacterial strains. The noticed selectiveness in the anti-fouling behavior of tested samples is presumably attributed to differences in the cell envelopes, the extent of elicit oxidative stress and soot-mediated formation of toxic ammonia molecules via hydrogen bonding. These unknown so far anti-bacterial mechanisms of the soot expand the horizons of practical applicability beyond the conventional framework and emphasize on its possible future use as a therapeutic agent.

1. Introduction

Since ancient times, the seas and oceans have been considered as a fundamental and limitless resource for the human civilization and planet Earth, because they regulate the climate, absorb heat and carbon dioxide, provide sustenance and generate ~70 % of oxygen, thus, helping to maintain livable atmospheric temperature [1,2]. With the tremendous advancement of shipbuilding in the Age of Discovery, between 15th–19th century CE, except for fisheries and maritime trade, the humanity has started to use the marine environments as a route for extensive overseas expeditions, most notable of which is the first wave of European colonization [3,4].

Nowadays, the global shipping activity is accompanied by inevitable transfer of invasive aquatic plants and animals, mainly through the ballast water of internal seawater systems and hull fouling [5–7]. Once attached to the ship's hull, these biospecies may move to a completely different geographic area and if survive the journey, they could establish a reproductive population and eradicate the native microorganisms – a process perceived as a significant threat to the oceans and coastal ecosystems [5]. On the other hand, from engineering point-of-view, the

biofouling of seawater delivery systems is detrimental to the proper engine cooling, freshwater production and air conditioning of the vessel [6]. Moreover, the accumulation of biomass corrodes the fuselage or as a minimum alters its hydrodynamic profile, by making the surface randomly roughened, increasing the solid-liquid frictional resistance and hence the fuel consumption and emissions of greenhouse gases [8–10]. Thus, inventing innovative strategies for combating the biofouling is a primary concern in environmental management, shipbuilding and marine transport industry [11].

As a spontaneous event, the biological colonization of a given solid surface immersed in the sea begins with the formation of a conditioning layer due to the adsorption of dissolved organic matter, including proteins, carbohydrates and other biomolecules [12]. Afterwards, the bacteria inhabiting the water attempt to irreversibly anchor to the surface via release of lipopolysaccharides and if this happens, they start proliferating and growing into colonies [12]. Further secretion of extracellular polymeric substances supports the strong adhesion of the cell colonies, serving as a platform for secondary colonization by diatoms, algae spores and larvae of invertebrates [12]. The settlement of such microfoulers creates a favorable environment for the aggregation of

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oysters, barnacles, seaweed and tubeworms, causing the formation of a macroscopic biofilm [12].

Since the aforementioned process initiates within minutes to hours upon immersion of the solid in water and lasts approximately a few weeks to months [12–14], the inceptive bacterial attachment on the seacraft might be avoided by means of self-polishing copolymers infused with tributyltin biocides that kill/inactivate the foulants [15–17]. Another possibility is to wait until sufficient amount of living matter accumulates on the surface and then remove it mechanically by the virtue of hydrodynamic forces exerted throughout the sailing [18–20]. Also, the employment of silver-doped materials is highly beneficial for elimination of the undesired bioadhesion, because the positively charged Ag^+ ions attract the negatively charged cell membrane and disrupt it, hence preventing the bacterial metabolism [21,22]. Furthermore, the Ag^+ ions are capable of destroying the proteins and genetic code of the bacteria, and thereby blocking their multiplication [21,23]. However, these techniques have insuperable shortcomings associated with risks of ecological catastrophe caused by the biocides [16,17], partial damage of the hull due to the mechanical cleaning and impossibility for complete biofilm detachment [19] or loss of Ag particles due to ultrasonication (for instance, if the material surface is treated with ultrasonically activated water stream [19]) or abrasion [21].

Taking into account the huge contemporary progress of physical, chemical and engineering sciences, the initial cell-solid surface interactions can be passively (without the aid of an external source) hindered using bioinert carbon soot coatings manipulating the dynamics of cell adhesion [24–27]. In particular, the work of bioadhesion approaches zero when the water contact angle (WCA) is close to 180° , establishing lower surface energy of the solid compared to the bacterial cell [28]. So, the soot's extreme water repellency ($\text{WCA} \geq 150^\circ$) determines very weak solid-cell contact and disposition of the bacteria in the domain of long-range attraction forces (>50 nm [29]), resulting in lack of biomass attachment [24]. Furthermore, the soot nanostructures impede the bacterial intercalation inside the pores, confirmed likewise for a variety of superhydrophobic materials [30,31], significantly reducing the amount of irreversibly adhered biospecies [25]. Also, the large fraction of air cavities along with the incorporation of silver nanoparticles in the structural matrix of the soot provokes bactericidal effect and physical injury of the bacteria due to their drooping among the surface protrusions [26,32]. Another recently presented artful strategy employs fluorinated candle soot coatings with near-infrared irradiation [33], benefiting from the repellence of the majority of microbes at early stage, similarly to other soot-based coatings [25], and ceasing the subsequent biofilm development via infrared-governed extermination of already bonded to the surface pathogens. Nevertheless, most of the studies concerning anti-microbial coatings are performed by static immersion in watery suspensions comprising individual bacterial strains cultured in a laboratory-controlled manner [20,24–27,30,31,33,34]. Only a limited number of them deal with submersion in an open sea, where the surface is influenced by underwater currents and attacked by a cocktail of marine microorganisms [35,36].

This article enriches the current state-of-the-art by demonstrating a pioneering research related to the anti-microbial behavior of different mechanically robust non-wettable carbon soot coatings directly plunged in a saltwater swamp. The collected experimental data yield new knowledge pertaining to the physicochemical mechanisms governing the anti-fouling capabilities of the soot in natural (out-of-lab) water environments. Such “real-life” tests elucidate also the extent to which the plastron (trapped air) layer formed between the micro/nanovalleys of the soot is preserved in dynamic conditions – a phenomenon that has not been investigated yet, but remaining one of the key challenges for the practical implementability of superhydrophobic surfaces [37,38].

2. Materials and methods

2.1. Materials

For the crafting of durable soot coatings, paper napkins and ethanol (99 wt%) were purchased from Mr. Bricolage and Valerus Ltd., Bulgaria. A liquefied cyanoacrylate glue (Stanger®) and fluorocarbon emulsion (Grangers Performance Proofer) were delivered by Office1 Superstore, Bulgaria, and Grangers International Ltd., UK, respectively. Medical Technics Engineering, Bulgaria, provided microscope glass slides, while Zoya Organic & Natural, Bulgaria, supplied the rapeseed oil used as a soot precursor.

2.2. Selection of soot patterns and coating deposition

The controlled combustion of rapeseed oil guarantees the production of different types of soot coatings [39], but the most appropriate for outdoor applications are those composed of weakly oxidized quasisquare-shaped carbon agglomerates [40] or prepared by combining textile products, cyanoacrylate glue and fluorocarbon solution [41]. In essence, a cylindrical paper wick soaked in rapeseed oil was ignited in a combustion chamber at incoming air flow of $0.0033 \text{ m}^3/\text{min}$. The nascent soot possessing quasisquare particle morphology was initially deposited on half the surface of a single 25×50 mm polished iron substrate and then entirely on a second iron plate, preliminary overlaid with a double-sided sticky tape and a paper napkin. *Sample1* (without a napkin) was dunked for 3 s in ethanol and 600 s in fluorocarbon, and then dried for 180 s on a hot plate at 100°C . *Sample2* (with a napkin) was processed with cyanoacrylate and fluorocarbon by following the technical protocol described in ref. [41]. These procedures ensured the presence of two functional coatings with sufficient mechanical robustness and different physicochemical profile, presumably affecting the anti-fouling activity [30–32]. A soot specimen with quasispherical morphology is excluded from this study, since one of the novel aspects herein is to identify the anti-microbial properties of a new generation ultradurable soot fabrics being a potential candidate for the shipbuilding industry. The second important aim that has not been accomplished hitherto is to gain new knowledge about the dynamics of biofouling at soot-metal interfaces instead of collating carbonaceous materials with similar anti-bioadhesiveness [25].

2.3. Geographic peculiarities of the sea coast accommodating the soot samples

The pair soot-coated iron substrates were submerged for seven days (August 6th–13th 2022) in a saltwater swamp having an average water temperature of 25°C in the summer at daily air temperature drifts within $15\text{--}35^\circ\text{C}$. It is located a few meters away from the seaside of Olympiaki Acti resort in the Thermaic Gulf of Katerini municipality, East Piera province, Greece. This aquatoria is considered as part of the Olympian Riviera, Aegean Sea, whose biochemistry is propitious to the bloom of more than sixty diverse harmful algae deteriorating the marine biota and water quality [42]. Plunging the samples in the nearby marsh instead of straight in the sea was justified due to several reasons: Most importantly, there was no suitable infrastructure (i.e., a private boat and calm bay) to host the coatings and avoid risk of human intervention (i.e., deliberate destruction of the specimens or stealing). Second, the swamp was very close to the waterfront (likely emerged via high tides), but meanwhile quite “cozy” for experimentation due to the slight denivelation with respect to the sea level (i.e., attracts less attention of the tourists). Third, the plethora of crabs, fish, eels, tadpoles and algae, as hinted in Fig. 1, resembled a case of a moored boat, whose hull is exposed to accelerated invasion of marine foulants [43]. The uncoated parts of iron substrates were used as a necessary control accounting the role of surface wettability and verifying whether the soot would act as an antibacterial agent towards the entire slab.



Fig. 1. A photograph of the swamp's flora/fauna and disposition of the samples. Nomenclature: *sample1* – iron substrate halfly coated with alcohol-fluorocarbon treated soot; *sample2* – soot fabrics-coated iron substrate.

2.4. Microbiological analysis

After one-week exposure to dynamic biofouling, the samples were moved into two sterile plastic containers filled with saltwater from the marsh and upon one-month storage in aerated dark/light conditions at $T \sim 20\text{--}27^\circ\text{C}$, the bacterial proliferation and biofilm growth were estimated. Initially, the soot-coated iron substrates were thoroughly washed with saline in order to get rid of the residual contaminated water. Afterwards, the specimens were dried, mounted in sterile Petri dishes and about 2 cm^2 of the soot-coated and bare iron surfaces were carefully peeled-off. Each substance (labelled as *soot1* – alcohol-fluorocarbon treated soot nanoparticles and bacteria scraped from *sample1*, *soot2* – soot fabrics and bacteria scraped from *sample2*, *uncoated1* – bacteria scraped from the uncoated part of *sample1* and *uncoated2* – bacteria scraped from the uncoated part of *sample2*) was placed in cuvettes containing 2 mL Dulbecco's Modified Eagle Medium/F12 (DMEM), 10 % fetal calf serum and a pH indicator phenol red [26]. Simultaneously, 0.2 mL of saltwater from both containers, marked as H_2O1 and H_2O2 , were diluted in 1.8 mL of the aforementioned buffer and all of the suspensions were incubated for 192 h at $T \sim 37^\circ\text{C}$.

In parallel, a set of six batch cultures were prepared by repeating the above steps and replacing the DMEM with distilled water, then subjected to serial dilution (1:10, 1:100, 1:1000 and 1:10000), seeded in Agar plates of five selective growth media (Candida Colorex, Sabouraud agar, Orientation agar Colorex, MacConkey agar and Columbia blood agar) via a cell spreader and incubated for 72 h at $T \sim 37^\circ\text{C}$.

The quantification of bacteria was performed by measuring the optical density (OD) of the subsequently centrifuged suspensions (to evade inherent measurement uncertainties induced by the cells) in the wavelength spectral range $\lambda \sim 320\text{--}700\text{ nm}$ through spectrophotometer (LAMBDA 1050, Perkin Elmer, USA). The OD values were correlated to the number of colony forming units per mL (CFU/mL), obtained by counting the colonies in the Agar media and multiplying by the dilution factor. The category of biospecies was defined based on the observed

morphology and staining, according to the nomenclature of Agar plates.

3. Results and discussion

3.1. Highlights of the physicochemical features of utilized soot coatings

Both types of flame deposited soot coatings are exhaustively analyzed via a set of sophisticated analytical instruments in previous research [40,41,44], so their physicochemistry is briefly introduced in Fig. 2 with the aim to avoid redundant repetition, but familiarize the reader with the basic features relevant solely to this study.

Although the generated pristine soot is the same (see Section 2.2), its post-synthesis chemical modifications yield two surfaces with contrasting pore size, ranging within 80–500 nm for the alcohol-fluorocarbon treated sample [44] and exceeding 300 μm for the napkin-coated one (see Fig. 2b). The ordinary quasisquare soot consists of carbon atoms covalently bonded to oxygen and fluorine functionalities [26,40], whereas the cross-linked filamentous structure of the soot fabric contains also nitrile, difluoromethyl and trifluoromethyl groups [41]. Furthermore, both soot patterns promote extreme droplet mobility (contact angle hysteresis within $1\text{--}5^\circ$), but have distinct water contact angles and durability towards abrasion [40,41]. This is a prerequisite for diverse anti-fouling capabilities especially in dynamic “real-life” conditions, where except by the surface profile, the bioadhesion would be controlled by the extent of mechanical wear.

3.2. Anti-bioadhesion behavior in dynamic saltwater environment

Several interesting results are documented at the end of the test period (37 days altogether), as illustrated in Fig. 3.

What can be seen is that both soot coatings are intact, but the entrapped air layer (plastron) is no longer available and the surface is fully wetted by the seawater, which opens a question whether the anti-bioadhesive properties of the soot are compromised or no. Furthermore,

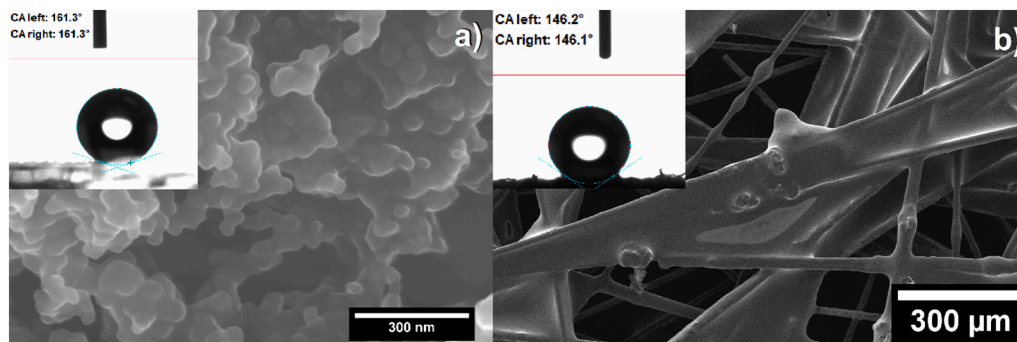


Fig. 2. Morphology and wettability (the inset images) of a) alcohol-fluorocarbon treated soot with quasisquare shape of the aggregates/agglomerates (*sample1*) and b) ultradurable soot fabrics (*sample2*) deposited on iron substrates. The scale bars look inconsistent, because panels a-b are taken at different magnifications of 50 kX and 100, respectively, due to the specificity of each surface.

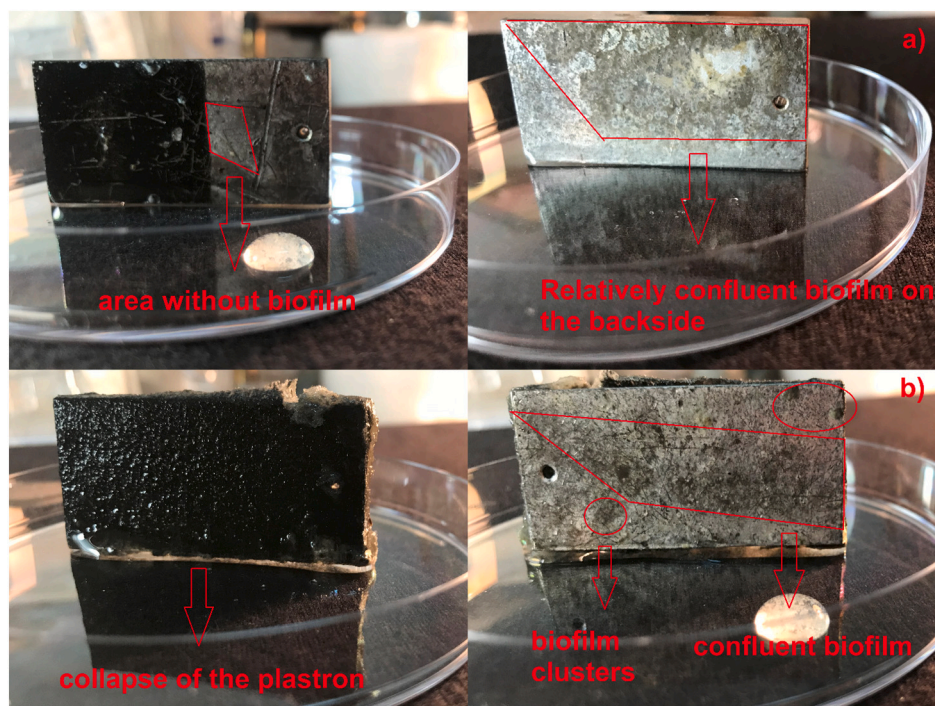


Fig. 3. A photograph of the front and backside of both iron substrates coated with a) alcohol-fluorocarbon treated soot (*sample1*) and b) ultradurable soot fabrics (*sample2*) after the end of the test period.

the backside of iron substrates is coated by visually notable confluent macroscopic biofilm, however, separate biomass clusters are recognizable onto the sample holding the soot fabric. Surprisingly, there is a region without biofilm at the *soot1*-metal boundary (see Fig. 3a), suggesting that this functional material strongly affects the kinetics of biofouling independently of the collapsed non-wettability.

To validate such a statement, the color changes of bacterial suspensions (see Section 2.4) are monitored on every 24 h, as shown in Fig. 4.

Knowing that the cell proliferation is accompanied by the production of organic acids, the increased acidity of the DMEM, expressed as a shift from bright pink to yellow color [45], is a clear sign for the presence of

bacteria. Within the first 24 h, all of the suspensions become yellow, but those carrying soot nanoparticles are darker i.e., characterized by higher pH and smaller amount of live bacteria (see Fig. S1 in SI) [45]. Upon prolonged incubation, the cuvette marked as *soot1* restores its pink color (see Fig. 4), implying bactericidal effect of the chemically modified soot with quasisquare morphology. It may be argued that such a recovery is inflicted by amendments in the cells' shape, accumulation of extracellular artifacts and secretion of ammonia [46,47], but if this hypothesis is correct, then the color restoration should be visible in all samples. Obviously, slight variation is discerned only in the *H₂O1* cuvette, likely due to leakage of ultrafine quasisquare soot aggregates in the water and killing some of the bacteria over the time interval of static assays (see

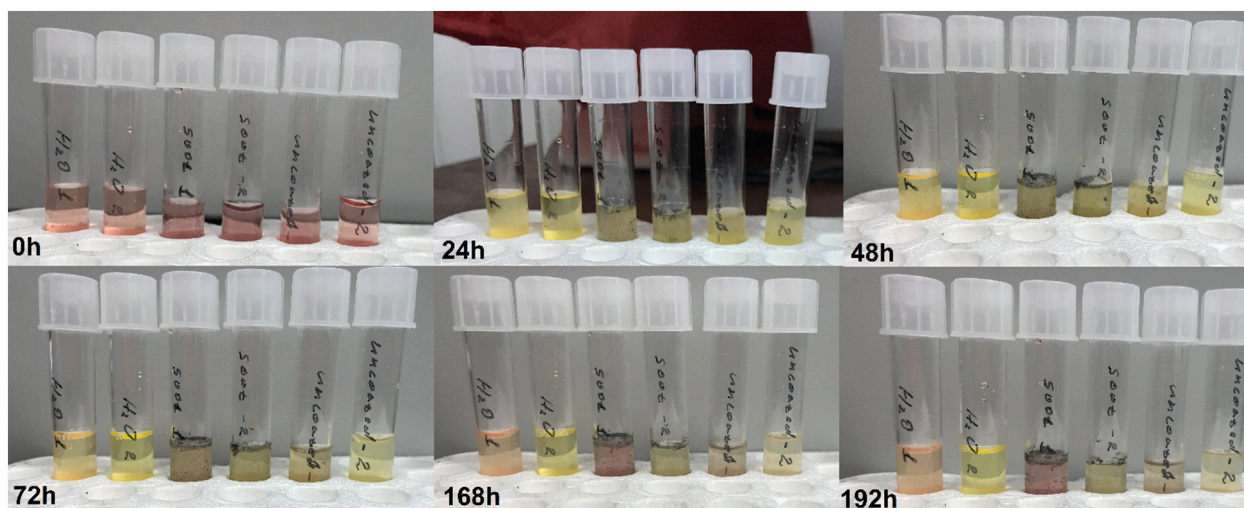


Fig. 4. Alterations in the pH of DMEM bacterial suspensions upon 192 h incubation. To differentiate the colors, the reader is kindly redirected to the online version of the article. Nomenclature: *soot1* – alcohol-fluorocarbon treated soot nanoparticles and bacteria scraped from *sample1*; *uncoated1* – bacteria scraped from the uncoated part of *sample1*; *soot2* – soot fabrics and bacteria scraped from *sample2*; *uncoated2* – bacteria scraped from the uncoated part of *sample2*; *H₂O1* – seawater from the container accommodating *sample1*; *H₂O2* – seawater from the container accommodating *sample2*.

Section 2.4), which is not feasible for the ultradurable soot fabric [41]. Although the assertion for probable bactericidal capabilities of the quasisquare-shaped soot agglomerates is speculative to a certain extent, it is clearly supported by the novel findings of other research groups worldwide, according to which, the insertion of different types of soot nanoparticles in biological suspensions causes the death of Gram-positive and Gram-negative bacterial strains such as *Staphylococcus aureus*, *Escherichia coli*, *Enterococcus faecalis*, *Pseudomonas aeruginosa* etc. [48–52].

The shift in pH of DMEM medium is further reaffirmed by spectrophotometric measurements, summarized in Fig. 5. The absorbance spectrum of pure DMEM/phenol red is described by two peaks, the first of which is centered at $\lambda \sim 415$ nm and accounts for the presence of large quantity of proteins, flavonoids, vitamins and nucleosides in the medium [53]. The other peak, located at $\lambda \sim 560$ nm, reflects the solution pH and in alkaline buffers (pH = 9) it is well-pronounced, but almost disappears if the pH decreases to 4.5 (see Fig. 4 in ref. [53]). In turn, the spectra recorded in Fig. 5 are in perfect agreement with the observed colorations, since the peak at $\lambda \sim 560$ nm diminishes for the most yellowish/acidic suspension (H_2O_2), but enhances for the pinkish/alkaline *soot1* duplicate.

Of particular practical significance is to clarify the category of bacterial strains being attached to the submerged substrates or prevailing primarily in the surrounding saltwater, which is feasible by analyzing the cell colonies depicted in Fig. 6. Five pathogenic Gram-negative (*Pseudomonas aeruginosa*, *Proteus vulgaris*, *Enterobacter aerogenes*, *Shigella sonnei* and *Escherichia coli*) and three Gram-positive (*Staphylococcus aureus*, *Enterococcus faecalis* and *Streptococcus pneumoniae*) bacteria, along with one Gram-positive fungus (*Candida albicans*), are localized in the Agar. The cells are unevenly distributed in the plates, revealing that there is a tendency in the microbial proliferation from sample-to-sample. For instance, *Streptococcus pneumoniae* (see the fifth Petri dish from left to right in each subpicture in Fig. 6) multiplies faster on the batches referring to the quasisquare soot (i.e., *soot1*, *uncoated1* and H_2O_1), whilst *Shigella sonnei* and *Escherichia coli* (see the fourth and fifth Petri dishes from left to right in each subpicture in Fig. 6) grow predominantly on the batches associated with the soot fabric (i.e., *soot2*, *uncoated2*, H_2O_2).

To quantify comprehensively the discerned trends of bacterial growth, the experimental data related to the number of CFU/mL and the pH of analyzed medium are presented in Table 1.

The thoroughly summarized experimental findings divulge suppressed multiplication of Gram-negative bacteria caused by the quasisquare soot and remarkably, this material endows the overall culture medium (the uncoated part and the water itself) with bacteriostatic

properties (see the data concerning *Pseudomonas aeruginosa*, *Proteus vulgaris*, *Shigella sonnei* and *Escherichia coli* at H_2O_1 and *uncoated1* in Table 1). In contrast, the soot fabric accelerates the proliferation of these bacteria, but impedes the reproduction of Gram-positive species (e.g., *Staphylococcus Aureus* and *Streptococcus pneumoniae*). Despite that, both soot specimens have smaller amount of attached cells compared to the uncoated backsides (juxtapose the total CFU/mL for *soot1-uncoated1* or *soot2-uncoated2* in Table 1), demonstrating potential for outdoor anti-microbial applications even upon loss of non-wettability. Last but not least, the spectrophotometrically measured pH conforms to the number of CFU/mL, validating again the colorations shown in Fig. 4.

3.3. Novel insights into the anti-microbial mechanisms of water-repellent rapeseed oil soot

The majority of ultra-water-repellent (also known as super-hydrophobic) coatings, including those made of carbon soot, sustain anti-biofouling ability due to the weak solid-liquid interfacial attraction, modifying the liquid's meniscus from concave to convex (in cases of full submersion instead of single droplets) and keeping the invading microorganisms away from the surface [24–26,29,33]. Specifically, a soot-layered solid would be free of any micro-/macroscopic biofouling as long as the coating maintains Cassie-Baxter wetting state [24,26]. If Cassie-to-Wenzel transition occurs (non-penetrating to liquid-penetrating regime), the custom topographical and morphological micropatterning starts governing the dynamics of bioadhesion [25,30–33], while the inclusion of an additional energy source (e.g., an infrared irradiator) may halt the biofilm growth [33]. From that fundamental point-of-view, the selective anti-microbial activity and the absence of adhered *Shigella sonnei* cells regardless of the collapsed water repellency (see Table 1) is pretty curious, inferring for other yet unexplored anti-fouling mechanisms of the soot, outlined below.

3.3.1. Importance of the bacterial cell envelope

All living organisms are composed of basic units called cells, whose principal distinction is in the complex multilayered structure of their wall [54]. Saying that, the cytoplasm of Gram-negative bacterial species is enclosed by a thin peptidoglycan cell wall and an outer membrane consisting of lipopolysaccharides, whereas the Gram-positive pathogens are surrounded merely of numerous and much thicker peptidoglycan layers [54].

One of the pathways for killing the bacteria considers electrostatic attraction of the cations of silver, gold, zinc, copper, etc., nanoparticles to the negatively charged molecules encompassing the cytoplasm of Gram-negative and Gram-positive bacteria, thus, disrupting the membrane and increasing its permeability [55]. Hence, inhibited enzyme activity and electrolyte imbalance occurs inside the cell, degrading the metabolic functions required to regulate the life-sustaining biochemical reactions.

Probably, the soot releases ions in the extracellular space, ruptures the cell envelope and trigger nuclear fragmentation or attachment to DNA molecules, impairing the biological processes in the cytoplasm [55]. Such an assumption might explain why the effect is much stronger for Gram-negative species, because their wall is thinner and probably more penetrable due to the lack of anionic glycopolymers (teichoic acid) [48]. It cannot, however, explicate why the anti-microbial effect is ascertained towards some Gram-positive bacteria too, especially for the sooted fabric (see Table 1). Moreover, the rapeseed oil soot has negative zeta potential [56], so the emerging electrostatic repulsion should significantly reduce the anti-biofouling activity, providing a space for scientific debates.

3.3.2. Oxidative stress-mediated cell inactivation

In multicellular systems, the imbalance among free radicals and antioxidants triggers oxidative stress originating from the accumulation of reactive oxygen species (ROS) such as superoxide anion, hydrogen

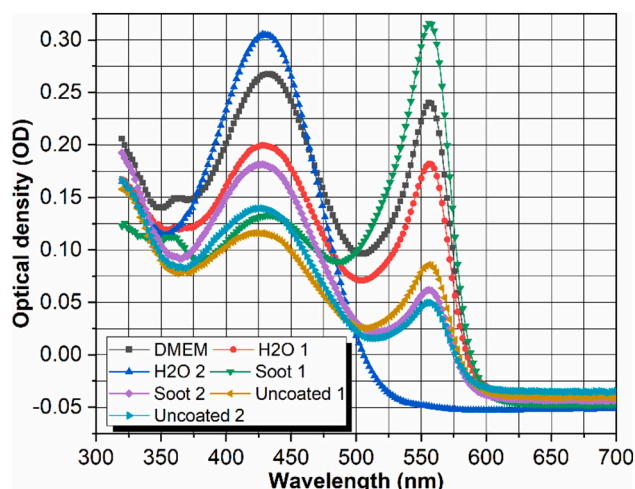


Fig. 5. Absorbance spectra (OD) of the bacterial suspensions.

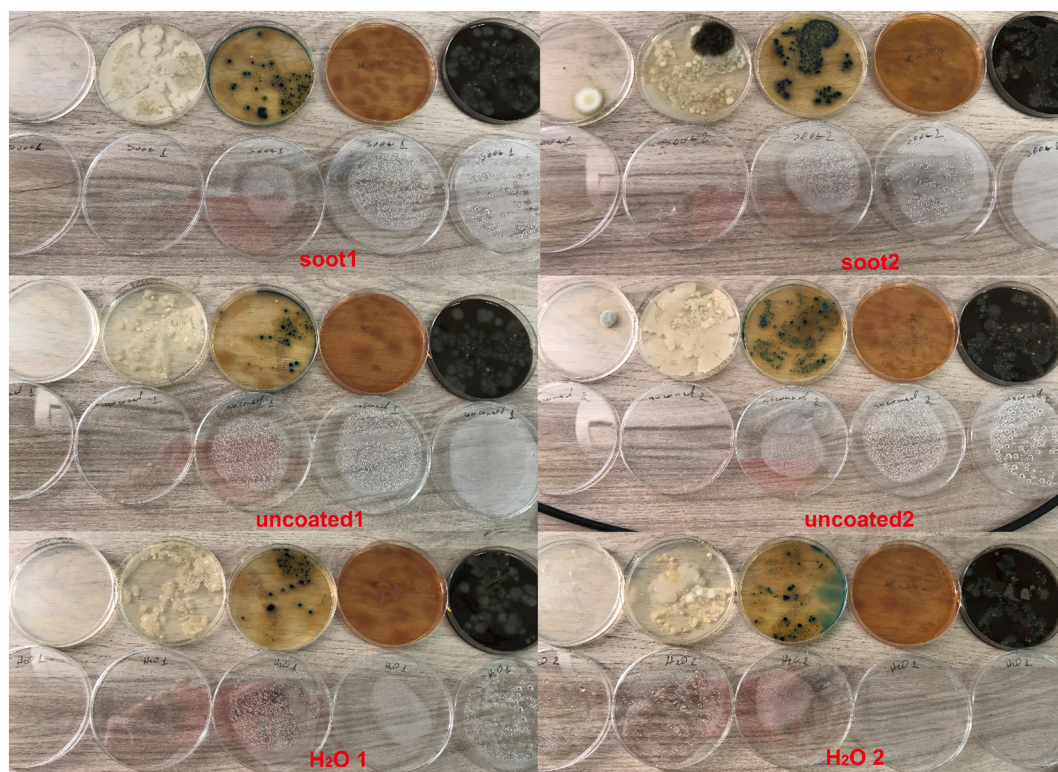


Fig. 6. Distribution of different bacterial strains and fungus in the selective growth media. For unambiguous categorization, enlarged version of the Petri dishes is available in Figs. S2–S7 in SI. Nomenclature: *soot1* – alcohol-fluorocarbon treated soot nanoparticles and bacteria scraped from *sample1*; *uncoated1* – bacteria scraped from the uncoated part of *sample1*; *soot2* – soot fabrics and bacteria scraped from *sample2*; *uncoated2* – bacteria scraped from the uncoated part of *sample2*; *H₂O1* – seawater from the container accommodating *sample1*; *H₂O2* – seawater from the container accommodating *sample2*.

Table 1

Number of colony forming units per mL (CFU/mL) as a function of the type and pH of the sample. The red and blue colors in the online version of the article indicate positive and negative effects towards the bacterial proliferation. The signs (+) and (–) denote the Gram-positive and Gram-negative classification, respectively.

Bacterial strain/Fungus	Type of sample											
	<i>H₂O1</i>		<i>H₂O2</i>		<i>Soot 1</i>		<i>Soot 2</i>		<i>Uncoated1</i>		<i>Uncoated2</i>	
	pH	CFU/mL	pH	CFU/mL	pH	CFU/mL	pH	CFU/mL	pH	CFU/mL	pH	CFU/mL
<i>Candida alb.</i> (+)		3×10 ⁵		4×10 ⁵		1×10 ⁵		3×10 ⁵		4×10 ⁵		4×10 ⁵
<i>Staphyl. Aureus</i> (+)		2×10 ⁵		2×10 ⁵		5×10 ⁵		1×10 ⁵		3×10 ⁵		5×10 ⁵
<i>Pseudomonas</i> (–)		2×10 ⁵		6×10 ⁵		1×10 ⁵		3×10 ⁵		3×10 ⁵		3×10 ⁵
<i>Proteus vulgaris</i> (–)		2×10 ⁵		5×10 ⁵		2×10 ⁵		4×10 ⁵		2×10 ⁵		3×10 ⁵
<i>Enter. Faecalis</i> (+)	6.7	1×10 ⁵	6	3×10 ⁵	7	1×10 ⁵	6.4	4×10 ⁵	6.5	1×10 ⁵	6.2	3×10 ⁵
<i>Enterobac. Aerog.</i> (–)		3×10 ⁵		3×10 ⁵		3×10 ⁵		3×10 ⁵		1×10 ⁵		1×10 ⁵
<i>Shigella sonnei</i> (–)		1×10 ⁵		1×10 ⁵		0		2×10 ⁵		0		3×10 ⁵
<i>Str. Pneumoniae</i> (+)		4×10 ⁵		3×10 ⁵		4×10 ⁵		1×10 ⁵		6×10 ⁵		2×10 ⁵
<i>E. coli</i> (–)		2×10 ⁵		3×10 ⁵		1×10 ⁵		3×10 ⁵		2×10 ⁵		4×10 ⁵
Total CFU/mL		2×10 ⁶		3×10 ⁶		1.8×10 ⁶		2.4×10 ⁶		2.2×10 ⁶		2.8×10 ⁶

peroxide, hydroxyl radical and singlet oxygen [55,57]. These ROS interrupt the physiological functionality of the cells via non-specific modifications of the critical amino acid residues in proteins, fatty acids in lipids and nucleic acids – a mechanism of nanotoxicity resulting in ceased biofilm development.

Since hydroxyl radicals are the most dangerous ROS capable of directly interacting with almost any living matter [57], and knowing that the rapeseed oil soot contains hydroxyl functional groups [26,56,58], it seems that the latter may induce proteolytic degradation, DNA damage and genomic instability in the bacteria, and subsequently their lethal end. Such a concept is credible, because it elucidates the previously unknown reason for the worse anti-bioadhesion performance

of the quasisquare soot compared to the quasispherical duplicate upon loss of superhydrophobicity (see Tables 3–4 in ref. [25]). Simply, the lower atomic percentage of OH groups in the former (see Fig. 4 in ref. [58]) is a prerequisite for weaker oxidative stress towards the targeted microorganisms.

3.3.3. Ammonia formation-controlled bactericidal performance

Some fungus and bacteria (e.g., *Candida albicans* and *Pseudomonas aeruginosa*) produce ammonia during their growth and morphogenesis [47,59], but this inorganic compound is found toxic to the development of other biospecies (e.g., *Escherichia coli* and *Enterobacter aerogenes*) [60].

From that starting point, we hypothesize that the increased amount

of elemental nitrogen due to the accumulation of living matter (see Fig. 6d in ref. [25]) would stimulate the formation of NH_3 molecules in the bacterial suspensions via hydrogen bonding to the hydroxyl and carboxyl sites of the soot. As a result, the lysozyme susceptibility increases, which cleaves the peptidoglycan component of the cell wall and alters the concentration of 2-keto 3-deoxyoctonate (KDO) in the outer membrane, leading to leakage of cytoplasm and cell death – a mechanism that is more efficient to Gram-negative bacteria [60]. In our opinion, that is why the quasisquare soot manifests bacteriostatic effect towards four out of five Gram-negative bacterial strains (see Table 1), but fails to do so with the Gram-positive species. Additionally, the soot fabric is mechanically ultradurable (it withstands abrasion and knife-scraping tests [41]), so the strengthened chemical bonds might be more non-reactive, hampering the nascence of hydrogen bonds.

3.4. Future outlook

Although our findings open many additional questions, the described anti-microbial pathways could be used as a groundwork for further exciting applications. For instance, understanding profoundly the biofilm-retarding properties of the rapeseed oil soot would be beneficial in tissue engineering, because it may help to accelerate the growth of eukaryotic or epithelial cells at the expense of harmful bacteria, identically to the well-known silver compounds [61]. Besides, the chemically functionalized soot introduced in this research is obviously appropriate for outdoor use in harsh conditions, but the eventual commercialization would be warranted if the non-wettable coating is designed to retain its air plastron for long periods of time. This could be feasible by fabricating nanoscale and hierarchically roughened surface topography (i.e., nanometer sized pillar-to-pillar distance and sub-micron/nanoprotrusions) for stabilizing the vapor phase of water [37] and mentioning that possibility, good pretenders are also the graphite derivatives such as graphene, expanded graphite, graphite nanosheets or carbon nanotubes [62]. In addition, if the ultradurable soot fabrics is capable of releasing heat upon absorption of infrared light, it could be adapted to the innovative anti-microbial technology presented by Lin et al. [33] in order to manufacturing an optimized ship hull composed of soot fabrics and embedded infrared irradiators. Such a design would ensure extended service life and acceptable anti-fouling activity even if Cassie-to-Wenzel interfacial transition occurs – existing drawbacks that still hinder the implementation of superhydrophobic materials in ship-building industry. Finally, since our research does not provide direct experimental evidences supporting the described anti-microbial mechanisms of the soot, due to currently insurmountable technical constraints, performing live/dead staining assay along with chemical analysis of the seawater is mandatory and planned as a future work.

4. Conclusions

The seven-days immersion in a seawater marsh of two polished iron substrates, whose front side was over-layered with durable quasisquare soot and a soot-based fabric, accordingly, triggered plastron depletion and declined water repellency. In spite of the compromised non-wettability, however, the coated parts of both substrates managed to selectively inhibit the multiplication of a variety of bacteria and fungi, and reduce the number of adhered cell colonies by a factor of 1.2–1.7 collated to the uncoated surfaces and buffer saltwater solutions. Since the main anti-microbial mechanism of superhydrophobic coatings, related to extended cell-solid separation distance and increased kinetic barrier needed to be overcome to initiate the microbial attachment [25,29], was no longer valid in the Wenzel state, the bacteriostatic activity of the soot was hypothetically attributed to proteolytic degradation, DNA damage, genomic instability and cleavage of the bacterial envelope, instigated by highly reactive hydroxyl radicals occupying the soot surface. The newly-discovered capacity of the soot coatings to mitigate the biofilm development even at fully wetted solid surface

provides fascinating opportunities for future employment of this material as a cell growth mediator in composite biological environments.

CRedit authorship contribution statement

Dr. Esmeryan conceived the idea to evaluate the anti-microbial activity of carbon soot coatings subjected to “real-life” highly-contaminated saltwater environment. He planned, designed and implemented the experiments, except the microbiological analysis. Dr. Esmeryan processed the raw data, contrived the scientific concept of the article and wrote its first and final versions. Mr. Chaushev performed the microbiological analysis and constructively reviewed the manuscript, whose scientific content was approved by the authors prior to submission.

Declaration of competing interest

The authors declare that they have not known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2023.107719>.

References

- [1] R. Costanza, The ecological, economic and social importance of the oceans, *Ecol. Econ.* 31 (1999) 199–213.
- [2] M. Visbeck, Ocean science research is key for a sustainable future, *Nat. Commun.* 9 (2018) 690.
- [3] C.R. Edwards, The impact of European overseas discoveries on ship design and construction during the sixteenth century, *GeoJournal* 26 (1992) 443–452.
- [4] P. Lawrence, The beginnings of European colonialism: a review article, *Oceania* 51 (1980) 53–59.
- [5] D. Nichol, Hull biofouling – old problem, new challenges, *Safety4Sea* (2019). <https://safety4sea.com/cm-hull-biofouling-old-problem-new-challenges/> (last access: 06.10.2022).
- [6] I. Davidson, P. Cahill, A. Hinz, D. Kluza, C. Scianni, E. Georgiades, A review of biofouling of ships' internal seawater systems, *Front. Mar. Sci.* 8 (2021), 761531.
- [7] F.T. Chan, D. Ogilvie, F. Sylvester, S.A. Bailey, Ship biofouling as a vector for non-indigenous aquatic species to Canadian arctic coastal ecosystems: a survey and modeling-based assessment, *Front. Mar. Sci.* 9 (2022), 808055.
- [8] M.P. Schultz, J.A. Bendick, E.R. Holm, W.M. Hertel, Economic impact of biofouling on a naval surface ship, *Biofouling* 27 (2011) 87–98.
- [9] O. Medina, J. Nocua, F. Mendoza, R. Gomez-Moreno, J. Avalos, C. Rodriguez, G. Morell, Bactericide and bacterial anti-adhesive properties of the nanocrystalline diamond surface, *Diam. Relat. Mater.* 22 (2012) 77–81.
- [10] A. Farkas, N. Degiuli, I. Martic, M. Vujanovic, Greenhouse gas emissions reduction potential by using antifouling coatings in maritime transport industry, *J. Clean. Prod.* 295 (2021), 126428.
- [11] Y. Liu, P. Guo, X. He, L. Li, A. Wang, H. Li, Developing transparent copper-doped diamond-like carbon films for marine antifouling applications, *Diam. Relat. Mater.* 69 (2016) 144–151.
- [12] A.E. Vellwock, H. Yao, Biomimetic and bioinspired surface topographies as a green strategy for combating biofouling: a review, *Bioinsp. Biomim.* 16 (2021), 041003.
- [13] S. Krishnan, R. Ayothi, A. Hexemer, J.A. Finlay, K.E. Sohn, R. Perry, C.K. Ober, E. J. Kramer, M.E. Callow, J.A. Callow, D.A. Fischer, Anti-biofouling properties of comblike block copolymers with amphiphilic side chains, *Langmuir* 22 (2006) 5075–5086.

- [14] E.T.H. Zürich, Surface Science and Technology, Reducing Biofouling With Polymer Brushes. <https://surface.mat.ethz.ch/research/polymer-brushes/reducing-biofouling-with-polymer-brushes.html> (last access: 07.10.2022).
- [15] T. Vladkova, Surface modification approach to control biofouling, in: H. C. Flemming, P.S. Murthy, R. Venkatesan, K. Cooksey (Eds.), Marine and Industrial Biofouling, Springer, Berlin, Heidelberg, 2009, pp. 135–163.
- [16] M. Ferrari, A. Benedetti, Superhydrophobic surfaces for applications in seawater, Adv. Colloid Interf. Sci. 222 (2015) 291–304.
- [17] M.S. Park, Y.D. Kim, B.M. Kim, Y.J. Kim, J.K. Kim, J.S. Rhee, Effects of antifouling biocides on molecular and biochemical defense system in the gill of the pacific oyster *Crassostrea gigas*, PLoS One 11 (2016) 0168978.
- [18] M.P. Schultz, C.J. Kavanagh, G.W. Swain, Hydrodynamic forces on barnacles: implications on detachment from fouling-release surfaces, Biofouling 13 (1999) 323–325.
- [19] M. Salta, L.R. Goodes, B.J. Maas, S.P. Dennington, T.J. Secker, T.G. Leighton, Bubbles versus biofilms: a novel method for the removal of marine biofilms attached on antifouling coatings using an ultrasonically activated water stream, Surf. Topogr. Metrol. Prop. 4 (2016), 034009.
- [20] K.Z. Hunsucker, H. Gardner, K. Lieberman, G. Swain, Using hydrodynamic testing to assess the performance of fouling control coatings, Ocean Eng. 194 (2019), 106677.
- [21] X. Wang, X. Ye, L. Zhang, Y. Shao, X. Zhou, M. Lu, C. Chu, F. Xue, J. Bai, Corrosion and antimicrobial behavior of stainless steel prepared by one-step electrodeposition of silver at the grain boundaries, Surf. Coat. Technol. 439 (2022), 128428.
- [22] Q. Liu, L. Ju, Z. Xin, D. Zan, C. Rong, Enhanced antibacterial activity and mechanism studies of Ag/Bi₂O₃ nanocomposites, Adv. Powder Technol. 29 (2018) 2082–2090.
- [23] J. Seyfi, M. Panahi-Sarmad, A.O. Ghodousi, V. Goodarzi, H.A. Khonakdar, A. Asefnejad, S. Shojaei, Antibacterial superhydrophobic polyvinyl chloride surfaces via the improved phase separation process using silver phosphate nanoparticles, Colloids Surf. B 183 (2019), 110438.
- [24] K.D. Esmeryan, C.E. Castano, M. Abolghasemibizaki, R. Mohammadi, An artful method for in-situ assessment of the anti-biofouling potential of various functional coatings using a quartz crystal microbalance, Sensors Actuators B Chem. 243 (2017) 910–918.
- [25] K.D. Esmeryan, I.A. Avramova, C.E. Castano, I.A. Ivanova, R. Mohammadi, E. I. Radeva, D.S. Stoyanova, T.G. Vladkova, Early stage anti-bioadhesion behavior of superhydrophobic soot based coatings towards *Pseudomonas putida*, Mater. Des. 160 (2018) 395–404.
- [26] K.D. Esmeryan, C.E. Castano, T.A. Chaushev, R. Mohammadi, T.G. Vladkova, Silver-doped superhydrophobic carbon soot coatings with enhanced wear resistance and anti-microbial performance, Colloids Surf. A Physicochem. Eng. Asp. 582 (2019), 123880.
- [27] C. Thamaraiselvan, E. Manderfeld, M.N. Kleinberg, A. Rosenhahn, C.J. Arnusch, Superhydrophobic candle soot as a low fouling stable coating on water treatment membrane feed spacers, ACS Appl. Bio Mater. 4 (2021) 4191–4200.
- [28] Y. Cui, W. Yuan, Thermodynamic modeling of algal cell–solid substrate interactions, Appl. Energy 112 (2013) 485–492.
- [29] S.H. Yoon, N. Rungraeng, W. Song, S. Jun, Superhydrophobic and superhydrophilic nanocomposite coatings for preventing *Escherichia coli* K-12 adhesion on food contact surface, J. Food Eng. 131 (2014) 135–141.
- [30] J. Bruzaud, J. Tarrade, E. Celia, T. Darmanin, E.T. de Givenchy, F. Guittard, J. M. Herry, M. Guilbaud, M.N. Bellon-Fontaine, The design of superhydrophobic stainless steel surfaces by controlling nanostructures: a key parameter to reduce the implantation of pathogenic bacteria, Mater. Sci. Eng. C 73 (2017) 40–47.
- [31] A.R. Mendez, T.Y. Tan, H.Y. Low, K.H. Otto, H. Tan, X. Khoo, Micro-textured films for reducing microbial colonization in a clinical setting, J. Hosp. Infect. 98 (2018) 83–89.
- [32] G.S. Watson, D.W. Green, J.A. Watson, Z. Zhou, X. Li, G.S.P. Cheung, M. Gellender, A simple model for binding and rupture of bacterial cells on nanopillar surfaces, Adv. Mater. Interfaces 6 (2019) 1801646.
- [33] Y. Lin, H. Zhang, Y. Zou, K. Lu, L. Li, Y. Wu, J. Cheng, Y. Zhang, H. Chen, Q. Yu, Superhydrophobic photothermal coatings based on candle soot for prevention of biofilm formation, J. Mater. Sci. Technol. 132 (2023) 18–26.
- [34] T.P. Rasitha, S. Sofia, B. Anandkumar, J. Philip, Long term antifouling performance of superhydrophobic surfaces in seawater environment: effect of substrate material, hierarchical surface feature and surface chemistry, Colloids Surf. A Physicochem. Eng. Asp. 647 (2022), 129194.
- [35] W.M.I.W.M. Kamaruzzaman, M.F.M. Fekeri, N.A.M. Nasir, N.A.S.M. Hamidi, M. Z. Baharom, et al., Anticorrosive and microbial inhibition performance of a coating loaded with *Andropogon paniculata* on stainless steel in seawater, Molecules 11 (2021) 3379.
- [36] P.P. Falara, N.D. Papadopoulos, P. Vourna, Microstructure and performance of antifouling coatings on high-strength steel substrates immersed in the marine environment, Micro 2 (2022) 277–294.
- [37] P.R. Jones, X. Hao, E.R. Cruz-Chu, K. Rykaczewski, K. Nandy, T.M. Schutzius, K. K. Varanasi, et al., Sustaining dry surfaces under water, Sci. Rep. 5 (2015) 12311.
- [38] F. Barghi, M. Entezari, S.F. Chini, A. Amirfazli, Effect of initial wetting state on plastron recovery through heating, Int. J. Heat Mass Transf. 156 (2020), 119705.
- [39] K.D. Esmeryan, C.E. Castano, Y.I. Fedchenko, R. Mohammadi, I.K. Miloushev, K. A. Temelkov, Adjustable optical transmittance of superhydrophobic carbon soot coatings by *in-situ* single-step control of their physicochemical profile, Colloids Surf. A Physicochem. Eng. Asp. 567 (2019) 325–333.
- [40] K.D. Esmeryan, A.H. Bressler, C.E. Castano, C.P. Fergusson, R. Mohammadi, Rational strategy for the atmospheric icing prevention based on chemically functionalized carbon soot coatings, Appl. Surf. Sci. 390 (2016) 452–460.
- [41] K.D. Esmeryan, Y.I. Fedchenko, S.D. Gyoshev, Y. Lazarov, T.A. Chaushev, T. Grakov, On the development of ultradurable extremely water-repellent and oleophobic soot-based fabrics with direct relevance to sperm cryopreservation, ACS Appl. Bio Mater. 5 (2022) 3519–3529.
- [42] L. Ignatiades, O. Gotsis-Skretas, A review on toxic and harmful algae in Greek coastal waters (E. Mediterranean Sea), Toxins 2 (2010) 1019–1037.
- [43] O. Brine, L. Hunt, M.J. Costello, Marine biofouling on recreational boats on swing moorings and berths, Manag. Biol. Inv. 4 (2013) 327–341.
- [44] K.D. Esmeryan, S. Vargas, S.D. Gyoshev, C.E. Castano, Water droplet bouncing on pre-frosted superhydrophobic carbon soot – a step forward in designing passive icephobic surfaces, Diam. Relat. Mater. 123 (2022), 108850.
- [45] Phenol red. https://en.wikipedia.org/wiki/Phenol_red#cite_note-Saha2015-2; (Last access: October 19th 2022).
- [46] A.J. Van Alst, R.M. LeVeque, N. Martin, V.J. DiRita, Growth Curves: Generating Growth Curves Using Colony Forming Units and Optical Density Measurements. <https://www.jove.com/v/10511/growth-curves-generating-growth-curves-using-colony-forming-units>; (Last access: October 19th 2022).
- [47] H.A. Danhof, S. Vylkova, E.M. Vesely, A.E. Ford, M. Gonzalez-Garay, M.C. Lorenz, Robust extracellular pH modulation by *Candida albicans* during growth in carboxylic acids, ASM mBio 7 (2016), e01646.
- [48] B. Mohanty, A.K. Verma, P. Claesson, H.B. Bohidar, Physical and anti-microbial characteristics of carbon nanoparticles prepared from lamp soot, Nanotechnology 18 (2007), 445102.
- [49] Habiba, I.U. Din, S. Tasleem, M. Mubashir, M.A. Bakht, Antimicrobial activity of antibiotic doped carbon nanoparticles extracted from kitchen soot against pathogenic Gram-negative bacteria, J. Pharma. Res. Int. 32 (2020) 63314.
- [50] V. Pahal, P. Kumar, R. Kumar, P. Kumar, V. Kumar, Characterization and comparison assessment of bactericidal activity of carbon nanodots (CDs) and nanoparticles (CNPs) prepared from soots of clarified butter and mustard oil, respectively, J. Appl. Biol. Biotechnol. (2022) (in press).
- [51] R.S. Kumar, B. Sampathkumar, N. Hemalatha, V. Suganthi, S. Bhuvanawari, N. K. Udaya Prakash, Antibacterial and antifungal studies on the biosoot of common weeds, J. Pharma. Neg. Results 13 (2022) 851–855.
- [52] N. Linson, J. Jacob, S. Kuriakose, Bovine serum albumin (BSA) stabilized wood soot based carbon nanoparticles as potentially viable anti-bacterial agents for biomedical applications, Braz. J. Chem. Eng. (2023), <https://doi.org/10.1007/s43153-023-00334-x> (in press).
- [53] P. Held, Using Phenol Red to Assess pH in Tissue Culture Media, BioTek, Application Note Live Cell Imaging, BioTek Instruments Inc., VT USA, 2018.
- [54] T.J. Silhavy, D. Kahne, S. Walker, The bacterial cell envelope, Cold Spring Harb. Perspect. Biol. 2 (2010), a000414.
- [55] N. Mammari, E. Lamouroux, A. Boudier, R.E. Duval, Current knowledge on the oxidative-stress-mediated antimicrobial properties of metal-based nanoparticles, Microorg. 10 (2022) 437.
- [56] K.D. Esmeryan, I. Rangelov, T.A. Chaushev, Hydrophobic soot nanoparticles as a non-cytotoxic motility activator of human spermatozoa, Nanoscale Adv. 4 (2022) 2806–2815.
- [57] R. Liu, Y. Bian, L. Liu, L. Liu, X. Liu, S. Ma, Molecular pathways associated with oxidative stress and their potential applications in radiotherapy, Int. J. Mol. Med. 49 (2022) 65.
- [58] K.D. Esmeryan, C.E. Castano, A.H. Bressler, M. Abolghasemibizaki, R. Mohammadi, Rapid synthesis of inherently robust and stable superhydrophobic carbon soot coatings, Appl. Surf. Sci. 369 (2016) 341–347.
- [59] G. Szita, G. Biro, A synthetic, selective culture medium for *Pseudomonas aeruginosa*, Acta Vet. Hung. 38 (1990) 187–194.
- [60] G. Naundorf, N.G. Aumen, Ammonia-induced cell envelope injury in *Escherichia coli* and *Enterobacter aerogenes*, Can. J. Microbiol. 36 (1990) 525–529.
- [61] T. Angelova, N. Rangelova, V. Uzunova, N. Georgieva, T. Andreeva, A. Momchilova, R. Tzoneva, R. Müller, Cytotoxicity and antibiofilm activity of SiO₂/cellulose derivative hybrid materials containing silver nanoparticles, Turk. J. Biol. 40 (2016) 1278–1288.
- [62] D.J. Babu, M. Mail, W. Barthlott, J.J. Schneider, Superhydrophobicity: superhydrophobic vertically aligned carbon nanotubes for biomimetic air retention under water (*Salvinia* effect), Adv. Mater. Interfaces 4 (2017) 1700273.